

Causality - SoSe 2025

do-Calculus

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In the last lecture, we discussed the following concepts:

- Confounding and Selection Bias
- Path Method for Linear SCMs
- Adjustment Sets
- Adjustment for Direct Causes
- Identifiability
- Back-Door Criterion

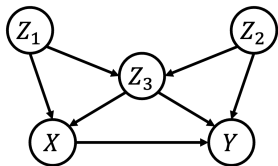
Definition Let $(\mathcal{G}, P)$ be a causal Bayesian network, $(i, k) \in \mathbf{V}, i \neq k$ (ordered). A set $\mathbf{Z} \subset \mathbf{V} \setminus \{i, k\}$ satisfies the **back-door criterion** relative to (i, k) in $\mathcal{G}$, if:

- $\mathbf{Z} \cap \text{De}(i) = \emptyset$, and
- $\mathbf{Z}$ blocks all **back-door paths** from i to k in $\mathcal{G}$, i.e., all paths between i and k that start with an arrow into i ($i \leftarrow \dots k$)

Theorem Let $(\mathcal{G}, P)$ be a CBN. If $\mathbf{Z} \subset \mathbf{V}$ satisfies the **back-door criterion** relative to (i, k) in $\mathcal{G}$, we have

$$p(x_k | do(x_i)) = \int_{\mathbf{x}_Z} p(x_k | x_i, \mathbf{x}_Z) p(\mathbf{x}_Z) d\mathbf{x}_Z$$

The back-door criterion is **sufficient** for adjustment.



Assume we have access to the CBN corresponding to the graph on the right.

- Which strategies can we use to compute the causal effect of X on Y ?
- Which adjustment sets fulfill the back-door criterion?
- Is there still a valid adjustment set when Z_1 is not observed?

In today's lecture, we will discuss *Pearl's do-calculus*. In particular, we discuss the following concepts:

- Simpson's Paradox (Revisited)
- Pearl's *do* calculus
- Front-Door Criterion
- Total- and Direct Causal Effects

Simpson's Paradox Revisited

Overall	
Treatment <i>a</i> : Open surgery	78% (273/350)
Treatment <i>b</i> : Percutaneous nephrolithotomy	83% (289/350)

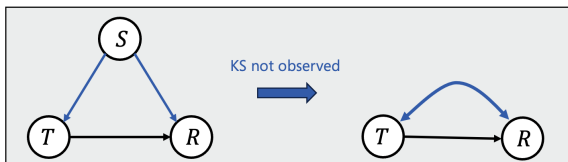
Figure 1: Simpson's paradox: Recovery rate after different treatments for kidney stones

Simpson's Paradox Revisited

	Overall	Patients with small stones	Patients with large stones
Treatment <i>a</i> : Open surgery	78% (273/350)	93% (81/87)	73% (192/263)
Treatment <i>b</i> : Percutaneous nephrolithotomy	83% (289/350)	87% (234/270)	69% (55/80)

Figure 2: Simpson's paradox: Recovery rate after different treatments for kidney stones

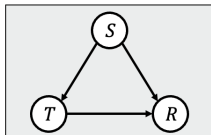
Causal view: (Unobserved) **confounding** through kidney stone size:



How can we compute the ACE
without interventions?

Simpson's Paradox Revisited

We know the **causal graph**, and we have **observational data**.



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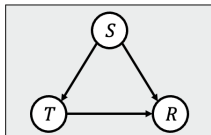
	Overall	Patients with small stones	Patients with large stones
Treatment a: Open surgery	78% (273/350)	93% (81/87)	73% (192/263)
Treatment b: Percutaneous nephrolithotomy	83% (289/350)	87% (234/270)	69% (55/80)

S satisfies the back-door criterion w.r.t. (T, R) !

We can compute the **post-interventional** probability via the back-door adjustment, i.e.,

$$p(R = 1 | do(T = t)) = \sum_s p(R = 1 | T = t, S = s)p(S = s)$$

Simpson's Paradox Revisited



&

	Overall	Patients with small stones	Patients with large stones
Treatment a: Open surgery	78% (273/350)	93% (81/87)	73% (192/263)
Treatment b: Percutaneous nephrolithotomy	83% (289/350)	87% (234/270)	69% (55/80)

Concretely:

$$\begin{aligned}
 p(R = 1 | do(T = a)) &= \sum_s p(R = 1 | T = a, S = s) p(S = s) \\
 &\approx 0.93 \frac{357}{700} + 0.73 \frac{343}{700} = \mathbf{0.832} \\
 p(R = 1 | do(T = b)) &\approx 0.87 \frac{357}{700} + 0.69 \frac{343}{700} = \mathbf{0.782}
 \end{aligned}$$

$$ACE(a, b) \approx 0.832 - 0.782 = 0.05$$

$$\neq p(R = 1 | T = a) - p(R = 1 | T = b) \approx -0.05$$

Simpson's Paradox Revisited

Question: What did we gain by the causal inference framework?

- We can clearly communicate our assumptions based on the DAG
- It is easy to discuss these assumptions with domain experts
- Given the DAG we know how to compute the causal effect

do Calculus

We know that we can *rewrite* $p(y|do(x))$ into probabilities that do not involve the *do* operator whenever we can find a **valid adjustment set**, which is e.g., guaranteed if the back-door criterion is satisfied.

Question: Is there a general framework to determine if we can *resolve* the *do* operator?

In other words: How can we determine whether a causal query is identifiable from observational data?

Goal of do-Calculus

In applications involving identifiability, the role of the do-calculus is to remove the do-operator from the query expression.

Quote Judea Pearl, Keynote at UAI 2021, "The Do-Calculus Revisited"

do-Calculus (Prerequisites)

Def. For a DAG $\mathcal{G}$ with nodes $\mathbf{V} = \{X, Y, Z, \dots\}$ we write:

- $\mathcal{G}_{\overline{X}}$ results from $\mathcal{G}$ by removing all **incoming** edges to X
- $\mathcal{G}_{\underline{X}}$ results from $\mathcal{G}$ by removing all **outgoing** edges from X

Examples from Pearl's book (2009):

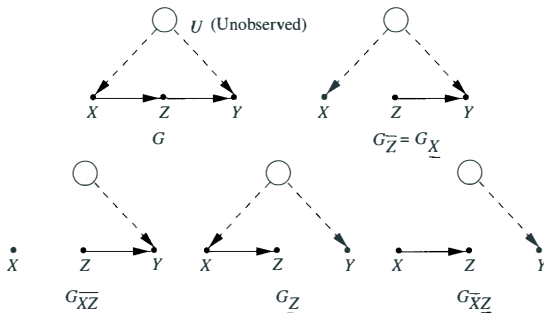


Figure 3.6 Subgraphs of G used in the derivation of causal effects.

do-Calculus (Rule 1)

Def. Let W, X, Y, Z^1 be disjoint subsets of $\mathbf{V}$ in a CBN $(\mathcal{G}, P)$.

Rule 1 (insertion/deletion of observations):

$$p(y|w, \mathbf{z}, do(x)) = p(y|w, do(x)) \quad \text{if} \quad Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{X}}} Z | \{W, X\}$$

i.e. d-sep. implies independence, also in a manipulated graph $\mathcal{G}_{\overline{X}}$.

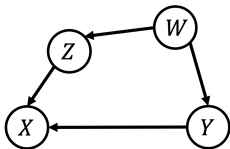
¹We omit bold face for these sets for visual purposes. We also omit this line in the following slides.

do-Calculus (Rule 1)

Rule 1 (insertion/deletion of observations):

$$p(y|w, \textcolor{blue}{z}, do(x)) = p(y|w, do(x)) \quad \text{if} \quad Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{X}}} Z | \{W, X\}$$

i.e. d-sep. implies independence, also in a manipulated graph $\mathcal{G}_{\overline{X}}$.



Query 1: $p(y|w, z, do(x)) = ?$

It holds that $Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{X}}} Z | \{W, X\}$, hence we can apply rule 1:

$$p(y|w, \textcolor{blue}{z}, do(x)) = p(y|w, do(x))$$

Note that we can also **insert variables** into the condition.

do-Calculus (Rule 2)

Rule 2 (action/observation exchange):

$$p(y|w, \text{do}(z), \text{do}(x)) = p(y|w, z, \text{do}(x)) \quad \text{if} \quad Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{XZ}}} Z | \{W, X\}$$

i.e. all back-doors from Z to Y in $\mathcal{G}_{\overline{X}}$ are blocked by $\{W, X\}$.

Examples:

Query 2: $\mathcal{G}$ is $X \rightarrow Y$, what is $p(y|\text{do}(x))$?

Since $Y \perp\!\!\!\perp_{\mathcal{G}_X} X | \emptyset$, we get $p(y|\text{do}(x)) = p(y|x)$

Query 3: $\mathcal{G}$ is $X \rightarrow Y$, what is $p(x|\text{do}(y))$?

$X \not\perp\!\!\!\perp_{\mathcal{G}_Y} Y$ ⚡ Not solvable so far...

Rule 2 (Examples)

Rule 2 (action/observation exchange):

$$p(y|w, do(z), do(x)) = p(y|w, z, do(x)) \quad \text{if} \quad Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{XZ}}} Z | \{W, X\}$$

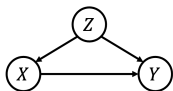
Query 4: $\mathcal{G}$ is $X \leftarrow U \rightarrow Y \leftarrow Z$, what is $p(z|do(x))$?

$$\begin{aligned} p(z|do(x)) &= p(z|x) && (\text{R2: } Z \perp\!\!\!\perp_{\mathcal{G}_X} X) \\ &= p(z) && (\text{simplified R1: } Z \perp\!\!\!\perp_{\mathcal{G}} X) \end{aligned}$$

Rule 2 (Examples)

Rule 2 (action/observation exchange):

$$p(y|w, \text{do}(z), \text{do}(x)) = p(y|w, z, \text{do}(x)) \quad \text{if} \quad Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{XZ}}} Z | \{W, X\}$$



Query 5: $p(y|\text{do}(x)) = ?$

$$Y \not\perp\!\!\!\perp_{\mathcal{G}_X} X \quad \text{⚡}$$

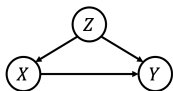
$$\begin{aligned} p(y|\text{do}(x)) &= \sum_z p(y, z|\text{do}(x)) \\ &= \sum_z p(y|\text{do}(x), z) p(z|\text{do}(x)) \quad (\text{factorization}) \\ &= \sum_z p(y|x, z) p(z|\text{do}(x)) \quad (\text{R2: } Y \perp\!\!\!\perp_{\mathcal{G}_X} X | Z) \end{aligned}$$

Rule 2 (Examples)

Rule 2 (action/observation exchange):

$$p(y|w, \text{do}(z), \text{do}(x)) = p(y|w, z, \text{do}(x)) \quad \text{if} \quad Y \perp\!\!\!\perp_{\mathcal{G}_{\underline{XZ}}} Z | \{W, X\}$$

Query 5: $p(y|\text{do}(x)) = ?$



$$Y \not\perp\!\!\!\perp_{\mathcal{G}_{\underline{X}}} X \quad \text{⚡}$$

$$\begin{aligned} p(y|\text{do}(x)) &= \sum_z p(y, z|\text{do}(x)) \\ &= \sum_z p(y|\text{do}(x), z) p(z|\text{do}(x)) \quad (\text{factorization}) \\ &= \sum_z p(y|x, z) \underbrace{p(z|\text{do}(x))}_{=?} \quad (\text{R2: } Y \perp\!\!\!\perp_{\mathcal{G}_{\underline{X}}} X | Z) \end{aligned}$$

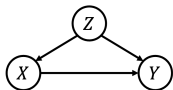
do-Calculus (Rule 3)

Rule 3 (insertion/deletion of actions):

$$p(y|w, \text{do}(z), \text{do}(x)) = p(y|w, \text{do}(x)) \text{ if } Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{XZ(W)}}} Z | \{W, X\}$$

where $Z(W) \subseteq Z$ that are not ancestors of an element of W in $\mathcal{G}_{\overline{X}}$.

Query 5 (continued):



$$\begin{aligned} p(y|\text{do}(x)) &= \sum_z p(y|\text{do}(x), z) p(z|\text{do}(x)) \\ &= \sum_z p(y|x, z) p(z|\text{do}(x)) \quad (\text{R2: } Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{X}}} X | Z) \\ &= \underbrace{\sum_z p(y|x, z) p(z)}_{\text{back-door adjustment}} \quad (\text{R3: } Z \perp\!\!\!\perp_{\mathcal{G}_{\overline{X}}} X) \end{aligned}$$

do-Calculus (Rule 3)

Rule 3 (insertion/deletion of actions):

$$p(y|w, \text{do}(z), \text{do}(x)) = p(y|w, \text{do}(x)) \text{ if } Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{XZ(W)}}} Z|\{W, X\}$$

where $Z(W) \subseteq Z$ that are not ancestors of an element of W in $\mathcal{G}_{\overline{X}}$.

Recall **Query 3**: $\mathcal{G}$ is $X \rightarrow Y$, what is $p(x|\text{do}(y))$?

$$X \not\perp\!\!\!\perp_{\mathcal{G}_Y} Y \quad \text{⚡}$$

However, we can apply Rule 3 since $X \perp\!\!\!\perp_{\mathcal{G}_{\overline{Y}}} Y$, and hence

$$p(x|\text{do}(y)) = p(x) \neq p(x|y)$$

The post-interventional query is clearly not equal to the conditional density.

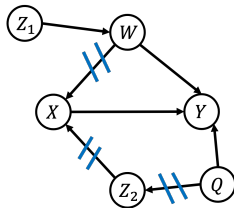
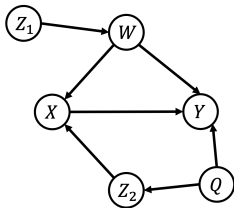
Rule 3 (Example)

Rule 3 (insertion/deletion of actions):

$$p(y|w, \text{do}(z), \text{do}(x)) = p(y|w, \text{do}(x)) \text{ if } Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{XZ(W)}}} Z|\{W, X\}$$

where $Z(W) \subseteq Z$ that are not ancestors of an element of W in $\mathcal{G}_{\overline{X}}$.

Query 5: $p(y|w, \text{do}(z), \text{do}(x))$?



We can apply Rule 3 here.

Def. Let W, X, Y, Z be disjoint subsets of $\mathbf{V}$ in a CBN $(\mathcal{G}, P)$.

Rule 1 (insertion/deletion of observations):

$$p(y|w, \textcolor{blue}{z}, do(x)) = p(y|w, do(x)) \quad \text{if} \quad Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{X}}} Z | \{W, X\}$$

i.e. d-sep. implies independence, also in a manipulated graph $\mathcal{G}_{\overline{X}}$.

Rule 2 (action/observation exchange):

$$p(y|w, do(\textcolor{blue}{z}), do(x)) = p(y|w, \textcolor{blue}{z}, do(x)) \quad \text{if} \quad Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{XZ}}} Z | \{W, X\}$$

i.e. all back-doors from Z to Y in $\mathcal{G}_{\overline{X}}$ are blocked by $\{W, X\}$.

Rule 3 (insertion/deletion of actions):

$$p(y|w, do(\textcolor{blue}{z}), do(x)) = p(y|w, do(x)) \quad \text{if} \quad Y \perp\!\!\!\perp_{\mathcal{G}_{\overline{XZ(W)}}} Z | \{W, X\}$$

where $Z(W) \subseteq Z$ that are not ancestors of an element of W in $\mathcal{G}_{\overline{X}}$.

Corollary 3.4.2 (in Pearl, 2009) A causal query q is **identifiable** in a model characterized by graph $\mathcal{G}$ if there **exists a finite sequence of transformations** (by Rules 1-3), that reduces q into a **standard probability expression** involving observable quantities.

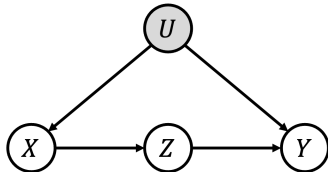
Further:

- The do-calculus has been shown to be **complete**, namely, sufficient to derive all identifiable causal effects (Shpitser and Pearl, 2006; Huang and Valtorta, 2006)

Front-Door Criterion

Front-Door Adjustment

Question: Can we identify $p(y|do(x))$ in the following graph?



Derivation in notes for Lecture 6!

Front-Door Criterion

Definition Let $(\mathcal{G}, P)$ be a causal Bayesian network, $(i, k) \in \mathbf{V}, i \neq k$ (ordered). A set $\mathbf{Z} \subset \mathbf{V} \setminus \{i, k\}$ satisfies the **front-door criterion** relative to (i, k) in $\mathcal{G}$, if:

- $\mathbf{Z}$ intercepts all directed paths from i to k (front-door paths),
- there is no (unblocked) back-door path from i to Z , and
- all back-door paths from Z to k are blocked by i .

Theorem Let $(\mathcal{G}, P)$ be a CBN. If $\mathbf{Z} \subset \mathbf{V}$ satisfies the **front-door criterion** relative to (i, k) in $\mathcal{G}$, we have²

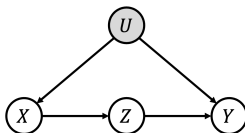
$$p(x_k | do(x_i)) = \int_{x_Z} p(x_Z | x_i) \int_{x'_i} p(x'_i) p(x_k | x'_i, x_Z) dx'_i dx_Z .$$

²Note that here $\mathbf{Z}$ is not a valid adjustment set.

Front-Door Criterion (Example)

Example:

- X : Smoking
- Y : Lung Cancer
- Z : Tar Deposits in Lungs
- U : Genes (Smoking Genes)



Assumptions:

- Genes do not influence the tar in the lungs (only the physical act of smoking influences tar deposits)
- There is no direct effect of smoking on cancer and no indirect path from smoking to cancer except the one through tar

Note: If we had measurements of X , Y , Z since the 1950s, we could apply the front-door adjustment under these assumptions.

Total- and Direct Causal Effects

UC Berkeley Admission Data³

Question: Is there a gender bias in college admission at UC Berkeley?

	All		Men		Women	
	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
Total	12,763	41%	8,442	44%	4,321	35%

Context:

- Study conducted in 1973 at the University of California, Berkeley
- Another classical example of Simpson's paradox

³Source Wikipedia (https://en.wikipedia.org/wiki/Simpson%27s_paradox)

UC Berkeley Admission Data

Question: Is there a gender bias in college admission at UC Berkeley?

Department	All		Men		Women	
	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
A	933	64%	825	62%	108	82%
B	585	63%	560	63%	25	68%
C	918	35%	325	37%	593	34%
D	792	34%	417	33%	375	35%
E	584	25%	191	28%	393	24%
F	714	6%	373	6%	341	7%
Total	4526	39%	2691	45%	1835	30%

Legend:

 greater percentage of successful applicants than the other gender

 greater number of applicants than the other gender

bold - the two 'most applied for' departments for each gender

Question: Is there a gender bias in college admission at UC Berkeley?

Recall:

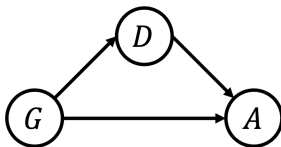
The **total average causal effect** is defined as

$$ACE(x, x') = E[Y \mid do(X = x)] - E[Y \mid do(X = x')] .$$

Question: Does computing the total average causal effect of *gender* on *admission* (if we could) help us answer the question above?

Causal View:

- Assume that Gender has an influence on the admission rate and the department selection
- The department also has an influence on the acceptance rate; here department is called a **mediator**
- In this case, we are only interested in the **direct effect** of G on A but not the mediated path $G \rightarrow D \rightarrow A$.



Controlled Direct Effect

The **controlled direct effect** of node i on node k is defined as

$$CDE = E[X_k \mid do(x_i+1), do(X_{Pa(k) \setminus \{i\}})] - E[X_k \mid do(x_i), do(X_{Pa(k) \setminus \{i\}})]$$

where we control for all remaining parents of k .

Intuition: We wiggle X_i while keeping all other parents fix to isolate the direct causal effect of X_i on X_k .

Question: How to set $do(X_{Pa(k) \setminus \{i\}})$?

- One could compute the CDE for all values in the support of $X_{Pa(k) \setminus \{i\}}$
- If $X_{Pa(k) \setminus \{i\}}$ is a **binary variable** it is common to compute CDE for both binary outcomes

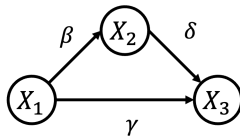
CDE (Example)

The **controlled direct effect** of node i on node k is defined as

$$CDE = E[X_k \mid do(x_i + 1), do(X_{Pa(k) \setminus \{i\}})] - E[X_k \mid do(x_i), do(X_{Pa(k) \setminus \{i\}})]$$

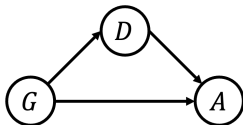
where we control for all remaining parents of k .

Example: Consider a linear SCM according to the CBN on the right



$$\begin{aligned} CDE &= E[X_3 \mid do(x_1 + 1), do(x_2)] - E[X_3 \mid do(x_1), do(x_2)] \\ &= ((x_1 + 1)\gamma + \delta x_2) - (x_1\gamma + \delta x_2) = \gamma \end{aligned}$$

If our assumed graph is correct, we could compute the **controlled direct effect** of gender on admission from observed data **for each department**.



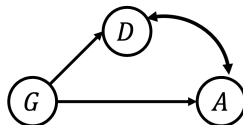
Since **none of the variables are confounded**, we could compute the CDE for department d as:

$$CDE(d) = E[A = 1 \mid G = \text{male}, D = d] - E[A = 1 \mid G = \text{female}, D = d]$$

Doing this calculation, we get that **4 out of 6 departments** are actually **discriminating against men**. But is this a fair assessment?

A Word of Warning

If this is the correct graph, we **cannot compute the CDE** from observations.



- Clearly, we **cannot resolve** the $do(D = d)$ as we cannot block all back-door paths between D and A
- Here, Kruskal correctly asked, “What if there is a confounder, State of Residence?” between department and admission and constructed a synthetic counter example (see *Book of Why*)

The problem above is generally problematic for mediation analysis and such back-doors should be carefully ruled out if possible.

How can we know?

Causal discovery might help...

- **Recommended reading** is Chapter 3.3 in *Causality* (Pearl, 2009)
- The example on lung cancer and the UC Berkeley admission problem are discussed in detail in Chapters 7 and 9, respectively in *The Book of Why* by Pearl and Mackenzie (2018).
- This course *builds upon and extends* the material from “*Causality*” (WiSe 2022/23) that was kindly provided by **Stefan Harmeling**.
- This lecture, *builds upon and extends* the slides of the course “*Causality*” (2021) by Christina Heinze-Deml.

Thank you!